1. Purpose

This experiment, the effect of the difference between masses on acceleration using the Atwoods machine will be observed and the average value will be found by collecting the data. Data will be collected from the LoggerLite program for the experiments carried out by keeping the total mass constant and changing the masses, and by keeping the difference between the masses constant and changing the masses.

2. Theory

A List of Materials

1. Motion Sensor
2. Rottery Motion Sensor
3. 2 Different Masses
4. Ruler
5. Mass Supports

In this experiment, the effect of changing masses on acceleration was examined using Newton's 2nd law, and the effect of changing the difference between masses on acceleration was examined. In the first stage, we carried out 3 different experiments by keeping the difference between masses constant and increasing the masses and recorded the results. In the second stage, we kept the total mass constant and increased the difference between the masses and recorded the results.

1. Figure 1: Atwoods Machine and Motion Detector

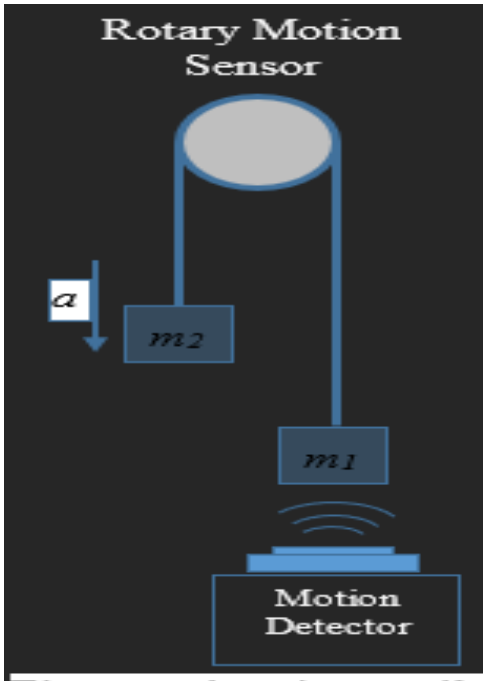

Figure 1 Atwood's machine and free-body diagram of the forces on each mass.

Part 1: Constant Mass Difference

We collected data from the LoggerLite program by adjusting the masses to m1 < m2 and performing 3 different experiments. Afterwards, we kept the difference between the masses constant and increased the masses equally and conducted 3 more experiments and collected the data.

Part 2: Constant Total Mass

In the second stage of the experiment, we determined the total mass so that m1 < m2, conducted 3 experiments and collected the data. Afterwards, we increased the difference between masses by keeping the total mass constant and conducted 3 more experiments. We averaged the acceleration values we obtained from the data we collected from the LoggerLite program.

Unlike the experiment we did in the last stage, we also conducted a Theoretical Experiment and recorded the percentage difference as the margin of error.